DiscoverX

User Manual

InCELL Detection Kit

Use with InCELL Hunter™ Cell Lines and InCELL Pulse™ Starter Kit Made with ProLabel® Vectors

Catalog No. 96-0079 Series

Please read this entire user manual before proceeding with the assay.
For additional information or Technical Support, see contact information below.

w discoverx.com usa 1.866.448.4864 e SupportUS@discoverx.com eu +44.121.260.6142 e SupportEurope@discoverx.com

© 2017 DiscoverX Corporation. All Rights Reserved. DiscoverX logo and DiscoverX are registered trademarks of DiscoverX Corporation. 20637 111017

Overview

The InCELL Detection Kit is to be used with InCELL Hunter™ cell lines (e.g. kinases, epigenetics, and protein binding cell lines) or the InCELL Pulse™ Target Engagement Starter Kit made with ProLabel® expression or cloning vectors. This kit has been validated for 96-well and 384-well plate formats, is easy-to-use, and results in a chemiluminescent signal that can be read with any standard plate reader.

The InCELL Hunter and Pulse platforms allow researchers to monitor compound-target engagement based on DiscoverX's proprietary enzyme fragment complementation (EFC) technology. This technology utilizes β-galactosidase split into two inactive fragments, an enhanced ProLabel (ePL) peptide (or enzyme donor) fragment and an enzyme acceptor (EA) fragment, that associate to form a fully active enzyme complex (complementation). Cells expressing the target protein are tested for changes in protein levels resulting from compound binding and altered protein stability.

Materials Provided

Catalog Number	**96-0079S**	**96-0079**	**96-0079L**
Number of Plates*	4 Plates	16 Plates	100 Plates
96-well, No. of Data Points	400	1,600	n/a
384-well, No. of Data Points	1,600	6,400	40,000
Kit Components (mL)			
EA Reagent	5	20	200
Lysis Buffer	5	20	200
Substrate Reagent	24	96	800
EA Dilution Buffer**	20	80	800
Total Kit Volume	**54**	**216**	**2,000**

* Plates are not included with kits

** EA Dilution Buffer is only needed for specific InCELL Hunter cell lines

Storage Conditions

Upon receipt, store reagents at -20°C. Thaw reagents at room temperature before use. Thawed reagents are stable for 4 days when stored at 2 to 8°C. The reagents can tolerate up to three freeze-thaw cycles with no impact on performance.

Additional Materials Needed

Required Materials	
InCELL Cells	
15 mL Polypropylene Tube (or larger, if necessary)	
Multimode or Luminescence Plate Reader	discoverx.com/instrument-compatibility
Disposable Reagent Reservoir	Thermo Scientific, Cat. No. 8094 or similar
Assay Plates	discoverx.com/microplates

Detection Assay Protocols

The procedures below detail the assay detection protocols to be used with InCELL Hunter™ cell lines and the InCELL Pulse™ Target Engagement Starter Kit. If running an assay using an InCELL Hunter cell line (e.g. an InCELL Hunter Kinase Cell Line), please refer to the InCELL Hunter user manual and the cell line-specific datasheet for assay preparation details and follow protocol A below. When using the InCELL Pulse Target Engagement Starter Kit, refer to the user manual for the InCELL Pulse Target Engagement Starter Kit for assay development guidelines and protocols and follow protocol B below.

Please refer to the specific datasheet for details on EA Reagent dilution and assay incubation time prior to starting this assay.

Follow the specific protocol below that best fits your application.

A. InCELL Hunter Cell Lines (384-Well Format) Detection Reagent Preparation

The following procedure details the assay detection step, in a 384-well format, for InCELL Hunter cell lines. Please refer to the cell line user manual for specific assay preparation details before proceeding.

Detection reagents must be prepared as a working solution prior to use. Dilution of the EA Reagent may or may not be required during preparation of this working solution; this requirement is specific for each InCELL Hunter cell line and is indicated on the cell line-specific datasheet.

Refer to specific InCELL Hunter Cell Line-specific datasheet for appropriate EA Reagent dilution requirement before proceeding.

1. If the InCELL Hunter cell line in use does not require dilution of the EA Reagent, then skip this step and proceed to Step 2. If EA dilution is necessary, then dilute the EA reagent by mixing 4 parts of EA Dilution Buffer with 1-part of EA Reagent in a separate tube.
2. Prepare a stock of working detection solution in a 15 mL polypropylene tube or reagent reservoir by mixing 1-part of EA Reagent (or diluted EA reagent), 1-part of Lysis Buffer, and 4 parts of Substrate Reagent according to the table below.

Do not substitute Lysis Buffer with alternative buffers. It is formulated for optimal assay performance.

The working detection solution should be prepared immediately before use (for assay detection).

Do not store pre-mixed working reagent for more than 24 hours.

Working Detection Solution for 384-Well Format		
Components	**Volume Ratio**	**Volume per Plate (mL)**
EA Reagent*	1	2
Lysis Buffer	1	2
Substrate Reagent	4	8
Total Volume		**12**

* Dilution of EA Reagent with EA Dilution Buffer may be required. Refer to Step 1 for instructions.

3. Add 30 µL of working detection solution to all wells of the assay plate.

4. Incubate the assay plate at room temperature, in the dark. The standard incubation period is 1 hour. Some cell lines may specify a different incubation period that could be as long as 16 hours (i.e. overnight). Please refer to the Functional Information section (i.e. the bottom of the first page) on the cell line-specific datasheet. Follow any alternative instructions that may be indicated for the cell line in use.

Working detection solution is light sensitive, thus incubation in the dark is necessary.

5. Refer to Reading the Assay Plate Section C.

B. InCELL Pulse™ (96-Well Format) Detection Reagent Preparation

The following procedure details the assay detection step, in a 96-well black PCR plate, for the InCELL Pulse Target Engagement Starter Kit.

Detection reagents must be prepared as a working solution during the compound incubation step of the assay. Dilution of the EA Reagent may or may not be required during preparation of this working solution. The optimal proportion of EA reagent required in the working solution must be determined during Phase I of the Pulse Assay Development. Please refer to your InCELL Pulse Target Engagement Starter Kit user manual for instructions on Phase I Assay Development, before proceeding.

Once an optimal formulation of working solution has been established for the assay being developed, it should be used for all subsequent experiments with that particular assay to minimize plate-to-plate, and day-to-day variation.

The working detection solution should be prepared immediately before use (for assay detection).

Do not substitute Lysis Buffer with alternative buffers. It is formulated for optimal assay performance.

1. Prepare a stock of working detection solution in a 15 mL polypropylene tube or reagent reservoir by mixing EA Reagent, Lysis Buffer, and Substrate Reagent. Use the optimal formulation of working detection solution determined during development of the InCELL Pulse assay for the target of interest. Refer to the Phase I Assay Development section in the user manual for the InCELL Pulse Target Engagement Starter Kit for details on determining the optimal formulation. The table below indicates three suggested formulations (i.e. EA-1, EA-3, and EA-10), each differing by the proportion of EA Reagent they contain (e.g. EA-1 contains 1 µL of EA Reagent, EA-3 contains 3 µL EA Reagent, etc).

Once an optimal working detection solution formulation is determined, use it for all experiments with the Pulse assay developed for the specific target

Working Detection Solution Recipes						
Working Detection Solution	**EA-1**		**EA-3**		**EA-10**	
	Volume (µL) per Well or Plate**					
Component	**Per Well**	**Per Plate**	**Per Well**	**Per Plate**	**Per Well**	**Per Plate**
EA Reagent	1	120	3	360	10	1200
Lysis Buffer	10	1200	10	1200	10	1200
Substrate Reagent	49	5880	47	5640	40	4800
Working Detection Solution (Total Volume*)	**60**	**7200**	**60**	**7200**	**60**	**7200**

* The total volume of EA Reagent plus Substrate Reagent (two of the three components included in the Working Detection Solution) should always equal 50 µL (out of the total 60 µL per well volume mix).

** Volumes account for overages and dead volumes

2. Add 60 µL of working detection solution to all wells of the assay plate.
3. Incubate your assay plate for 1 hour at room temperature in the dark.
4. Refer to Reading the Assay Plate Section C

Working detection solution is light sensitive, thus incubation in the dark is necessary.

C. Reading the Assay Plate

The following procedure details the reading of samples from Sections A or B above.

1. Read samples on a standard luminescence plate reader at 0.1 to 1 second per well for photomultiplier tube readers or 5-10 seconds for imager. The actual signal characteristics over time are affected by lab conditions, such as temperature. The user should establish an optimal read time. Luminescence readout collects signal from all wavelengths. Some instrument manufacturers may include a cutoff filter at high wavelengths, but usually no wavelength setting is needed for luminescence readout.
2. Data analysis can be performed using your choice of statistical analysis software (e.g. GraphPad Prism, Molecular Devices Softmax Pro, Biotek Instruments Gen5, Microsoft Excel, etc.).

For additional information or technical support, see contact information below.

Limited Use License Agreement

These products may be covered by issued US and/or foreign patents, patent application and subject to Limited Use Label License.

Please visit discoverx.com/license for limited use label license and trademark information for this product.